Day 14 Problem Set

Geometry Summer Credit | Trig Modeling in Context

Name:	Date:	Period:

Directions

Draw a labeled right triangle for each context. Write a trig equation, solve, and include units. Round decimal answers to the nearest tenth.

Support Practice: Read the Model

1.	Define angle of elevation in your own words.

2.	Define angle of depression in your own words.

3.	A person looks up at a roof. From the person's horizontal line, is the viewing angle elevation or depression?

4.	A pilot looks down at a runway. From the pilot's horizontal line, is the viewing angle elevation or depression?

5.	A vertical height and horizontal distance are used. Which trig ratio connects them?

Core Practice: Heights and Distances

6.	A student stands 25 ft from a tree. The angle of elevation to the top is 40 degrees. Find the tree height. Assume eye level is at the ground.

7.	A building casts a 48-ft shadow when the angle of elevation of the sun is 37 degrees. Find the building height.

8.	From the top of a 70-ft cliff, the angle of depression to a boat is 24 degrees. Find the horizontal distance to the boat.

9.	A kite is 65 ft above the ground. The angle of elevation from a student is 52 degrees. Find the horizontal distance to the point directly below the kite.

10.	A ramp rises 4 ft and makes an 8-degree angle with the ground. Find the ramp length.

11.	A 30-ft support cable makes a 63-degree angle with level ground. How high up a pole does it attach?

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Continued | Multi-Step Models and Spiral Review

12.	A person with an eye level of 5 ft stands 32 ft from a light pole. The angle of elevation to the top is 47 degrees. Find the total pole height.

13.	From a balcony 45 ft above the ground, the angle of depression to a parked car is 35 degrees. Find the horizontal distance to the car.

14.	A surveyor stands 60 ft from a building. The angle of elevation to the roof is 29 degrees. The instrument is 4.5 ft above the ground. Find the building height.

15.	Two observers stand on the same straight path from a tower. The closer observer is 30 ft from the tower and measures an angle of elevation of 61 degrees. Find the tower height. Then find the angle of elevation for an observer 55 ft from the tower.

16.	A drone is 120 ft above the ground. Its angle of depression to a landing marker is 41 degrees. Find the direct line-of-sight distance to the marker.

17.	Error analysis: A student finds the triangle height in an eye-level problem but reports it as the object's total height. Explain and correct the process.

18.	Spiral review: Find the missing side if a right triangle has legs 9 and 12.

19.	Spiral review: A 30-60-90 triangle has short leg 7. Find the long leg and hypotenuse.

20.	Spiral review: A right triangle has opposite side 8 and hypotenuse 17. Find the acute reference angle.

21.	Challenge: Explain why keeping full calculator precision until the last step improves a multi-step model.